Math 427, Fall 2026. Homework 5

(1) **[2.6] 11.** Let H and K be subgroups of some group G.
   (a) Show that $Ha \cap Ka = (H \cap K)a$ for all $a \in G$.
   (b) Given $a, b \in G$, show that either $Ha \cap Kb$ is empty or $Ha \cap Kb = (H \cap K)c$ for some $c \in G$.

(2) **[2.6] 30.** Let G be any group (possibly infinite) that has no subgroups except $\{1\}$ and G. If $|G| \geq 2$, show that G is finite and cyclic and that $|G|$ is prime. (Converse of Corollary 3 of Lagrange's theorem.)

(3) **[2.6] 31.** Let $K \subseteq H \subseteq G$ be groups. Show that both $|G : H|$ and $|H : K|$ are finite if and only if $|G : K|$ is finite, and then $|G : K| = |G : H||H : K|$.
[*Hint*: If $|H : K| = n$, let $Kh_1, Kh_2, \ldots, Kh_n$ be the distinct cosets of K in H. Show that $Hg = Kh_1g \cup Kh_2g \cup \cdots \cup Kh_ng$ is a disjoint union for all $g \in G$.]

(4) **[2.6] 32.** Let H and K be subgroups of a group G with $|G : H| = m$ and $|G : K| = n$.
   (a) Show that $|G : H \cap K| \leq mn$. [*Hint*: $(H \cap K)g = Hg \cap Kg$ for all $g \in G$.]
   (b) If $\gcd(m, n) = 1$, show that $|G : H \cap K| = mn$. [*Hint*: Exercise 31.]

(5) **[2.6] 34.** Show that A_4 has no subgroup of order 6, and hence that the converse of Lagrange's theorem is false. [*Hint*: Theorem 3.]

(6) **[8.3] 13.** Let $G = (\mathbb{R}, +)$ and define $a \cdot z = e^{ia}z$ for all $z \in \mathbb{C}$ and $a \in G$. Show that $\mathbb{C}$ is a G-set, describe the action geometrically, and find all orbits and stabilizers.

(7) **[8.3] 25.** Let X be a G-set, let H be a subgroup of G, and let $x \in X$. Show that:
   (a) H acts on the orbit $G \cdot x$ by $h \cdot (a \cdot x) = (ha) \cdot x$ for all $h \in H$ and $a \cdot x \in G \cdot x$.
   (b) If $H \triangleleft G$, the orbits of H in $G \cdot x$ all have the same cardinality.

(8) **[2.8] 12.** Let $K \triangleleft G$ where K is cyclic. Show that every subgroup of K is normal in G.

(9) **[2.8] 24.** A subgroup H of a group G is called a **characteristic subgroup** if $\sigma(H) \subseteq H$ for all automorphisms σ of G.

(a) Show that every characteristic subgroup is normal.

(b) Show that if H is characteristic in G then $\sigma(H) = H$ for all $\sigma \in \operatorname{aut} G$.

(c) If $G = C_2 \times C_2$ show that $H = C_2 \times \{1\}$ is normal in G but not characteristic. [*Hint*: Consider $\sigma : G \to G$ given by $\sigma(x, y) = (y, x)$.]

(d) Show that the center $Z(G)$ is characteristic in G.

(e) If $H \subseteq K \triangleleft G$ and H is characteristic in K, show that $H \triangleleft G$.

(f) If K is characteristic in H and H is characteristic in G, show that K is characteristic in G. (Compare with Exercise 4.)

(g) Show that every subgroup of a cyclic group G is characteristic in G. Is this true if G is merely abelian?

(h) If H and K are characteristic in G, show that $H \cap K$ is characteristic in G.

(i) If H is a subgroup of G, let $K = \{g \in G \mid g \in \sigma(H) \text{ for all } \sigma \in \operatorname{aut} G\}$. Show that K is characteristic in G, that $K \subseteq H$, and that K contains every characteristic subgroup of G that is contained in H.

(10) **[2.8] 25.** If X is a nonempty subset of a group G, define the **normalizer** $N(X)$ of X by

$$N(X) = \{a \in G \mid aXa^{-1} = X\}.$$

(b) If H is a subgroup of G, show that $H \triangleleft N(H)$.

(c) If H is a subgroup of G, show that $N(H)$ is the largest subgroup of G in which H is normal. That is, if $H \triangleleft K$, and K is a subgroup of G, then $K \subseteq N(H)$.

(11) **[2.8] 28.** If X is a nonempty subset of a group G, define the **centralizer** $C(X)$ of X by $C(X) = \{c \in G \mid cx = xc \text{ for all } x \in X\}$. Note that $C(G) = Z(G)$.

(b) If $K \triangleleft G$, show that $C(K) \triangleleft G$.

(12) **[2.9] 1.** In each case find the cosets in G/K, write down the Cayley table of G/K, and describe the group G/K.

(a) $G = D_6$ and $K = Z(D_6)$

(b) $G = Q$ and $K = Z(Q)$

(c) $G = A \times B$, A and B arbitrary groups, and $K = \{(a, 1) \mid a \in A\}$

(d) $G = \langle a \rangle \times \langle b \rangle$, where $o(a) = 8$ and $o(b) = 2$, and $K = \langle (a^2, b) \rangle$

(13) **[2.9] 17.** If G is abelian, let $T(G)$ denote the set of elements in G of finite order.

(a) Show that $T(G)$ is a subgroup of G—the **torsion subgroup**.

(b) Call G a **torsion-free group** if $T(G) = \{1\}$. Show that $G/T(G)$ is torsion free.

(c) Call G a **torsion group** if $T(G) = G$. If H is a subgroup of G, show that G is a torsion group if and only if both H and G/H are torsion groups.